

OMAR SAMI FAHMI ASKAR

Computer Science Student | AI Developer | Full-Stack Web Developer

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PROFESSIONAL SUMMARY

Computer Science student and AI developer with hands-on experience across the full stack — React.js, Node.js, Python, and SQL/PostgreSQL — and applied machine learning through prompt engineering and AI-focused coursework and training (NTI AI for Business, Microsoft/MCIT AI Literacy, Google 'Build with AI', Dubai Centre for AI '1 Million Prompts'). Co-founder of Nexera, a software and AI startup delivering client websites, e-commerce stores, and custom software. Combines practical software engineering with a growing AI specialization, targeting roles in AI development, machine learning, or full-stack engineering.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), SQL

AI & Prompt Engineering: Machine Learning fundamentals, Prompt Engineering, AI Literacy, AI for Business

Frontend: HTML5, CSS3, React.js, Responsive Web Design

Backend & APIs: Node.js, FastAPI, REST APIs

Databases: SQL, PostgreSQL, Supabase, Firebase, MongoDB

Tools & Platforms: Git, GitHub, Vercel, Railway

Professional Skills: Leadership, Communication, Teamwork, Problem Solving

EXPERIENCE

Co-Founder — Nexera (thenexera.tech)

July 2026 — Present

- Co-founded and helped launch Nexera, a software and AI startup building websites, Shopify stores, and custom software systems for clients.
- Led end-to-end product development across frontend and backend, from requirements gathering through deployment.
- Managed client communication and delivery, translating business requirements into functional, responsive digital products.

PROJECTS

Breast Cancer AI Prediction SaaS | github.com/omarsamy-web/breast_cancer_prediction

React.js, Vite, Tailwind CSS, Node.js, Express, FastAPI, scikit-learn, MongoDB, Supabase, JWT, Railway

- Architected and built a full-stack SaaS platform for AI-powered breast cancer diagnosis prediction, structured as three independent services: a React + Vite + Tailwind frontend dashboard, a Node.js + Express REST API backend, and a FastAPI + scikit-learn ML microservice.
- Implemented a multi-model training pipeline supporting KNN, SVM, Decision Tree, and Random Forest, with automatic model selection based on a combined F1 + accuracy score.
- Trained on a 1-million-row dataset with 40 numeric features, enabling high-confidence predictions via a medical feature form interface.
- Built a complete SaaS billing layer with Free, Pro (\$29), and Clinic (\$99) tiered quotas for predictions, trainings, and dataset uploads; enforced server-side with HTTP 402 quota-exceeded responses and JWT-secured endpoints.
- Designed for production deployment on Railway with separate services for frontend, backend, and ML layer; supports both MongoDB Atlas and Supabase for persistence.

Multi-Tenant AI Voice Ordering Platform | github.com/omarsamy-web/restaurant

Python, FastAPI, SQLite (PostgreSQL-ready), OpenAI Whisper, Twilio, pytest, HMAC Webhooks

- Developed a multi-tenant SaaS AI voice agent that answers restaurant phone calls, handles natural-language order conversations, authenticates customers, and submits orders to restaurant POS systems—with deterministic pricing and automatic human handoff for edge cases.

- Engineered a dual-brain agent architecture: an OpenAI function-calling LLM brain for production and a deterministic rule-based NLU state machine for offline development, testing, and evaluation—both sharing a single guarded tool executor.
- Achieved 46/46 unit and integration tests passing and 14/14 evaluation scenarios passing (~92% average score), covering complex modifications, authentication lockouts, payment declines, and kitchen rejections.
- Performed load testing: 200/200 voice-order pipeline executions with 0 errors, p50 latency $\approx$ 1.36s, throughput $\approx$ 5.3 orders/sec; raw database write path benchmarked at 66 orders/sec on SQLite.
- Implemented a comprehensive security model: PBKDF2-hashed PINs/OTPs, server-side pricing engine (no LLM arithmetic), authorization policy engine for large orders, 3-attempt auth lockout, and HMAC-signed webhooks with exponential backoff retry logic.
- Built a Restaurant API v1 with scoped, revocable API keys (sk_test_* / sk_live_*), rate limiting, idempotency keys, and IP allowlisting; designed the 36-table relational schema for zero-friction migration to PostgreSQL/Supabase.

Personal Portfolio | github.com/omarsamy-web/portfolio

HTML, CSS, JavaScript

- Designed and deployed a responsive personal portfolio website showcasing projects, certifications, and technical skills.

EDUCATION

Bachelor of Science in Computer Science (In Progress)

Menoufia National University — Egypt | 2023 — Expected 2027

CERTIFICATIONS & TRAINING

- **AI for Business** — NTI & ITIDA, 120 hours, Score: 87.5% — Jul–Aug 2026
- **AI Literacy** — AI Empower Yourself by MCIT, Microsoft & Fast Lane — Jun 2026
- **1 Million Prompts** — Dubai Centre for Artificial Intelligence & Dubai Future Foundation
- **Build with AI: Masr Edition** — Google for Developers × Information Technology Institute (ITI)
- **Python Lab Workshop** — Microsoft Student Ambassadors
- **CIB Certificate** — Jul–Aug 2026

ADDITIONAL INFORMATION

Events & Communities: AI Is Everything Conference — Participant, 2025 (AI, machine learning, and generative AI sessions; networking with developers and startup founders)

Languages: Arabic (Native), English (Good), Italian (Basics)